UrbanLens — Team Backlog

Everything left to finish the app, split five ways and sequenced by what blocks what. Each task has an acceptance test, so “done” is not a matter of opinion. Built from the PRD conformance audit at commit **1991d15**.

A note on the split

You asked for five people but named six. I have written the frontend as **three independent groups** (FE-A map and layers, FE-B panels and workflows, FE-C quality and access). If Harshil, Krish and Dhruvya are three people, take one group each. If frontend is really two people, fold FE-C into the other two — the groups share no files, which is why they are cut this way.

1. How to work through this

Rule	Why
One task = one branch = one PR	Keeps review small and lets two people work the same area without conflict
`npm run demo` and `npm run verify:ui` must pass before merge	These are the only safety net; a red harness means the demo is broken
Never merge a task whose acceptance test you cannot run	If it cannot be checked it is not finished
Data tasks land first	Rudra sits on the critical path — see §2. Anushka and FE-A are blocked until the real layers exist
Anything modelled must say so in the API response	PRD §30 and §70. This is the project's strongest quality; do not erode it

2. The critical path

Read this before picking anything up. Two people are gated on Rudra, and one item is gated on nobody and must happen first.

```
M0  DEMO-SAFE (this week)
    FE-A1 remove fake NDVI/UHI  ---+
    VD-1  deploy engine            +--> a demo you can show anyone
    FE-B1 deploy web           ---+

M1  REAL OBSERVATION (2-3 weeks)
    RD-1 GHSL built-up (2 dates) -----> AN-1 real temporal model
    RD-2 Sentinel-2 + indices     -----> FE-A2 honest NDVI layer
                                  -----> AN-4 land-cover segmentation
    RD-3 Copernicus DEM           -----> VD-4 real slope + flood risk

M2  SCALE + FORECAST (4-6 weeks)
    VD-2 PostGIS  ---> VD-3 vector tiles ---> FE-A4 tiled parcel layer
    AN-1 --------------------------------> FE-B4 forecast UI

M3  PRODUCT POLISH (parallel, any time)
    FE-B2 compare view   FE-B3 report export   FE-C* quality
```

Do FE-A1 today

Two map layers are labelled “Vegetation & NDVI Canopy” and “Urban Heat Island (UHI)” but are computed from a radial distance formula. It is the only place the product claims something it does not do, and it is fifteen minutes of work. Until it is fixed, do not enable those layers in front of a judge.

3. Rudra — Data & Sourcing

You are the critical path. Anushka cannot build a real forecast and the frontend cannot show honest vegetation until your layers land. Everything you produce goes into `refined/` or `web/data/engine/` with a provenance note that the API surfaces.

ID	Task	Effort	Blocked by	Done when
RD-1	**GHSL built-up rasters for two dates** (e.g. 1990 + 2020). Clip to each study area, reduce to per-parcel built-up fraction, write a history file. PRD §35	2 days	—	`/api/growth` reports built-up from observed data, and the response's source field says “GHSL”, not “modelled”
RD-2	**Sentinel-2 composite + NDVI / NDBI / NDWI.** Cloud-free seasonal composite, compute the three indices, aggregate per parcel and to a grid. PRD §31	3 days	—	Per-parcel vegetation_percent and water_percent come from imagery; a grid layer of real NDVI exists for FE-A2
RD-3	**Copernicus DEM** — elevation and derived slope per parcel. PRD §36	1 day	—	Parcel elevation_m is sampled, not modelled; slope available as a new field
RD-4	**Water bodies layer** from OSM natural=water + NDWI. PRD §7 lists it and it is missing	0.5 day	RD-2	A water layer renders on the map and environmental scoring uses real water proximity
RD-5	**Ward-level Census PCA** — replace the area × road-density population model with real ward counts where published. PRD §34	2 days	—	Population source flips from “derived” to “census” for wards that have real counts; the rest stay labelled derived
RD-6	**AUDA development plan / zoning** — georeference the DP sheets if obtainable. PRD §37	3 days	—	Zoning conflicts become real findings rather than method demonstrations; the caveat text is removed
RD-7	**Bhuvan LULC** as an independent cross-check on land use. PRD §32	1 day	—	A comparison report showing agreement rate between OSM land-use tags and Bhuvan LULC
RD-8	**Provenance manifest.** One machine-readable file recording every layer's source, licence, fetch date and known limits; serve it from `/api/layers`	1 day	—	Every layer in the UI can show where it came from and when it was fetched

Order: RD-1 then RD-2 (they unblock other people), then RD-3, then the rest. RD-6 is the highest-risk item — start asking AUDA early, because it may simply not be obtainable, and if so that is a finding worth stating rather than a task worth chasing.

4. Ved — Backend & Spatial Engine

You own the engine and the API contract. Two themes: make the substitutions the audit flagged defensible or unnecessary, and prepare for more than one city.

ID	Task	Effort	Blocked by	Done when
VD-1	**Deploy the engine** — container + Cloud Run or Railway, env-configured CORS origin. PRD §47	0.5 day	—	A public URL answers `/api/health` and the deployed web app uses it
VD-2	**PostGIS migration** — schema per PRD §48–55, SQLAlchemy models, loader script from the current GeoJSON. Keep the in-memory path working as a fallback	5 days	VD-1	Every route returns identical figures with the DB backend; `npm run demo` passes against both

ID	Task	Effort	Blocked by	Done when
VD-3	**Vector tiles + bbox queries** for parcels. The current `/api/parcels` returns everything in one response — about 290 MB at state scale	3 days	VD-2	The map requests only the visible extent; initial payload for Ahmedabad drops below 500 KB
VD-4	**Real flood risk and slope** from the DEM, replacing the modelled elevation	1 day	RD-3	Parcel flood_risk derives from elevation and water proximity; the "modelled" label is removed for it
VD-5	**OSRM routing** for the 15-minute analyzer, replacing straight-line distance. PRD §45	1.5 days	VD-1	Accessibility uses real drive/walk time; the response states which engine produced it
VD-6	**Vectorise the population rasteriser.** It builds one shapely Point per cell in a Python loop; `STRtree.query(points, predicate)` is about 25x faster (measured)	0.5 day	—	Grid build for the metro region drops from ~0.8 s to under 0.05 s; population total still conserved exactly
VD-7	**Unit tests** for scoring, raster and parcel enrichment. Two end-to-end harnesses exist; there is no unit layer	2 days	—	pytest covers decay_score, norm, final_score, population conservation and nearest-facility lookups; CI runs it
VD-8	**Precompute pipeline** — move parcel enrichment offline into a cache artefact rather than startup warming	2 days	VD-2	Engine startup is instant and enrichment is a build step, not a runtime cost
VD-9	**Report generation endpoint** — a parcel or site-selection result as a PDF the planner can file. PRD §46 mentions generated reports	2 days	—	`POST /api/report` returns a PDF with the scores, the explanation and the provenance

5. Anushka — Machine Learning

The current model is honest but limited: it scores *development pressure*, not a dated forecast, because the repo has no observed built-up extent at two dates. RD-1 removes that constraint. Until it lands, work on evaluation and explainability — both are things judges probe and neither is blocked.

ID	Task	Effort	Blocked by	Done when
AN-1	**Real temporal growth model.** Label = "was open in year A, built-up in year B" from the GHSL pair. The feature pipeline is already correct — only the label changes. PRD §11	3 days	RD-1	The model predicts urbanisation between two observed dates; the caveat in development_model.py is rewritten, and the layer can honestly be called a forecast
AN-2	**Baseline comparison.** Logistic regression and Random Forest against the same split, reported side by side. PRD §11 names all three	1 day	—	A table of accuracy / ROC AUC / CV for all three models, with a written justification for the one shipped
AN-3	**Per-prediction explanations (SHAP).** Feature importance is global; a planner asking "why is *this* cell high risk?" needs a local answer. PRD §20	2 days	—	`/api/growth/prediction` can return the top contributing features for a given cell

ID	Task	Effort	Blocked by	Done when
AN-4	**Land-cover segmentation on Sentinel-2.** The one place PRD §44 permits PyTorch — built-up / vegetation / water / bare from imagery	5 days	RD-2	A land-cover raster produced by the model, with accuracy reported against a held-out sample
AN-5	**Hyperparameter tuning + calibration.** Grid or Optuna search; check predicted probabilities are calibrated, since they are shown to users as percentages	1.5 days	—	A calibration curve in the model report; tuned parameters committed with the metrics that justify them
AN-6	**Model card.** One page: intended use, training data, metrics, limitations, what would make it wrong	0.5 day	AN-1	Committed at `docs/model-card.md` and linked from the growth panel
AN-7	**Retraining CLI + drift note.** Make retraining reproducible for a new city and record what changes when the input data does	1 day	—	`python -m app.ml.train <city>` documented, and a note on how metrics vary across the four areas

Start with AN-2 and AN-5. They need no new data, they harden the answers to the questions judges actually ask, and AN-2 produces the comparison table the technical reference currently promises in prose.

6. Frontend A — Map, layers & visual honesty

Suggested owner: Harshil. Files: `components/map/`, `config/layers.ts`, `lib/mapdata.ts`

ID	Task	Effort	Blocked by	Done when
FE-A1	**Remove or relabel the fake NDVI and UHI layers.** They are radial formulas presented as satellite products. PRD §70	15 min	—	Either the layers are gone, or their label and legend say “illustrative — not satellite-derived” and the API says the same
FE-A2	**Real NDVI / vegetation layer** from the imagery pipeline, with a legend showing the actual index range	1 day	RD-2	The layer renders values from Sentinel-2 and its provenance chip reads “Sentinel-2”
FE-A3	**Water bodies + flood-risk layers** on the map. PRD §7 lists both	0.5 day	RD-4, VD-4	Both layers toggle, have legends, and are used by the environmental constraint panel
FE-A4	**Switch the parcel layer to vector tiles** and load by viewport	1.5 days	VD-3	Panning loads tiles on demand; time-to-interactive on first load improves measurably
FE-A5	**Satellite basemap comparison / swipe** between two years. PRD §61 asks for satellite comparison	1.5 days	RD-2	A swipe or side-by-side control compares two imagery dates over the same extent
FE-A6	**Map legend completeness.** Every active layer needs a legend entry; several heat layers currently render with no scale shown	0.5 day	—	Toggling any layer shows what its colours mean, including units
FE-A7	**Map performance pass** — verify layer add/remove does not leak sources, and that switching study area fully tears down the previous one	1 day	—	Switching areas ten times does not grow memory or duplicate map sources

7. Frontend B — Panels, workflows & reporting

Suggested owner: Krish. Files: `components/panels/`, `components/parcels/`, `services/`

ID	Task	Effort	Blocked by	Done when
FE-B1	**Deploy the web app** to Vercel, pointed at the deployed engine. PRD §47	0.5 day	VD-1	A public URL loads the product and every panel shows real figures
FE-B2	**Compare candidates view.** PRD §64 lists “compare candidates” in the site-selection workflow and it does not exist	2 days	—	Two or three candidates can be shown side by side with their factor breakdowns aligned for comparison
FE-B3	**Export a recommendation as PDF** — the parcel, its scores, the explanation and the provenance, as a filed document	1.5 days	VD-9	A button on the parcel drawer and on a site-selection result produces a PDF
FE-B4	**Forecast UI** once the model is temporal — year selector, probability bands, and wording that matches what the model actually claims	1.5 days	AN-1	The growth panel says “predicted urbanisation by <year>” only when the model supports it
FE-B5	**Per-cell explanation popover** on the prediction layer, using SHAP output	1 day	AN-3	Clicking a prediction cell shows the top factors behind its score
FE-B6	**Scenario save / recall.** PRD §55 defines a planning_scenarios table; the simulator currently forgets everything on reload	2 days	VD-2	A simulation can be named, saved and reopened with its before/after intact
FE-B7	**Empty and error states audit.** Every panel should say what went wrong and what to do, never sit on a spinner	1 day	—	With the engine stopped, every panel shows an actionable message; no panel spins forever
FE-B8	**Copilot streaming + suggested follow-ups** so answers feel immediate and the planner learns what else they can ask	1.5 days	—	Tokens stream as they arrive and each answer offers two relevant follow-up questions

8. Frontend C — Quality, access & demo readiness

Suggested owner: Dhruvya. If frontend is only two people, split this between A and B. Files: `app/`, `components/layout/`, `scripts/verify-ui.mjs`

ID	Task	Effort	Blocked by	Done when
FE-C1	**Guided demo mode.** A scripted walkthrough of the PRD §74 story that a presenter can step through without hunting for controls	2 days	—	A “Demo” control steps through all 12 beats of the story, each step landing on the right panel and map view
FE-C2	**Keyboard and screen-reader pass.** The map canvas currently has no ARIA role or label and no keyboard path; panels are reachable but unlabelled	2 days	—	Every interactive control is reachable by keyboard with a visible focus ring; the map has a role, a label and keyboard pan/zoom
FE-C3	**Responsive layout.** The shell has zero breakpoints — fixed 336 px panels and absolute positioning. It is unusable below about 1200 px	3 days	—	The product is usable on a 1024 px laptop and degrades sensibly on a tablet

ID	Task	Effort	Blocked by	Done when
FE-C4	**Extend verify-ui.mjs** to cover the new surfaces as they land — compare view, export, scenario save, demo mode	ongoin g	—	Every new feature ships with an assertion in the harness; the run stays green
FE-C5	**Loading and transition polish.** Skeletons that match final layout, so panels do not jump when data arrives. PRD §73 phase 10	1 day	—	No layout shift between skeleton and loaded state
FE-C6	**Error boundary + offline state.** A thrown render error currently takes the whole product to a blank screen	0.5 day	—	A component error shows a recoverable panel, not a white page; engine-unreachable is stated plainly
FE-C7	**Onboarding for a first-time planner** — a short overlay explaining the six modes	1 day	—	First visit explains the rail; dismissible and remembered

9. Milestones

Milestone	Contains	Ready when
M0 — Demo-safe this week	FE-A1, VD-1, FE-B1, plus a full rehearsal of the §74 story	A public URL shows the whole story, nothing on screen claims more than it does, and both harnesses are green
M1 — Real observation 2–3 weeks	RD-1, RD-2, RD-3, RD-4, VD-4, VD-6, FE-A2, FE-A3, AN-2, AN-5	Built-up history, vegetation and elevation all come from observed data. “Modelled” disappears from four layers
M2 — Forecast + scale 4–6 weeks	AN-1, AN-3, AN-6, VD-2, VD-3, VD-8, FE-A4, FE-B4, FE-B5	The 2030 layer is a genuine dated forecast, and the app serves more than one city without a 290 MB payload
M3 — Product parallel	FE-B2, FE-B3, FE-B6, FE-C*, VD-7, VD-9, RD-5..8, AN-4, AN-7	Compare, export, save, accessible, responsive, tested

10. If you only finish three things

Priority	Task	Why this one
1	FE-A1 — remove the fake NDVI/UHI layers	It is the only dishonest thing in an otherwise scrupulous product, and it takes fifteen minutes. Everything else you build is worth less if a judge finds this first
2	VD-1 + FE-B1 — deploy	A live URL removes every “works on my machine” risk and lets judges explore after you leave the room
3	RD-1 + AN-1 — GHSL and the temporal model	This converts your headline ML claim from “development pressure” to a real dated forecast. It is the single biggest technical credibility gain available, and it is two people for about a week

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